

Optimal Chemo Treatment Times

The constraint is $x_1 + a_2 x_2 = 1$ (total dose 1) and we minimise $x_1 + x_2$. By the Fundamental Theorem of Linear Programming, the optimum lies at a “corner” of the feasible region ¹. Here the only corner solutions are $(x_1, x_2) = (1, 0)$ or $(0, 1/a_2)$. If $a_2 < 1$ then $1/a_2 > 1$ so $(1, 0)$ gives the smaller time (1) and is optimal; if $a_2 > 1$, $(0, 1/a_2)$ is best.

From the code one finds

$$f(n) = \sum_{k=3}^{n+2} \frac{1}{2^k} + (\text{negligible terms}),$$

a finite geometric series. Using the sum formula (with first term $a=1/8$ and ratio $r=1/2$) we get $f(n) \approx \frac{1/8(1-(1/2)^{n+2})}{1-1/2} = \frac{1}{4}(1-(1/2)^{n+2})$, so $a_2 = 2f(n) \approx 1 - (1/2)^{n+2}$ (plus an extremely tiny positive correction). In any case $a_2 < 1$ (even if $n \rightarrow \infty$, $a_2 \rightarrow 1$ from below) ². Thus the faster rate is $a_1 = 1$, and the time should all come from the first therapy.

Hence the optimal solution is $x_1 = 1, x_2 = 0$ which in two-decimal form is $x_1 = 1.00$, $x_2 = 0.00$. (Any deviation increases $x_1 + x_2$.) This choice delivers dose 1 in total time 1, which is minimal ¹.

Answer: $x_1 \approx 1.00$, $x_2 \approx 0.00$. (Only the first therapy is used, since $a_2 < 1$.) ¹ ²

¹ 5.6 - Linear Programming

<https://people.richland.edu/james/lecture/m116/systems/linear.html>

² Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series